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Keeping a weather eye on prediction markets: The influence of environmental conditions on forecasting accuracy

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ABSTRACT

Increasingly, prediction markets are being embraced as a mechanism for eliciting and aggregating dispersed information and providing a means of deriving probabilistic forecasts of future uncertain events. The efficient market hypothesis postulates that prediction market prices should incorporate all information that is relevant to the performances of the contracts traded. This paper shows that such may not be the case in relation to information regarding environmental factors such as the weather and atmospheric conditions. In the context of horserace betting markets, we demonstrate that even after the effects of these factors on the contestants (horses and jockeys) have been discounted, the accuracy of the probabilities derived from market prices is affected systematically by the prevailing weather and atmospheric conditions. We show that significantly better forecasts can be derived from prediction markets if we correct for this phenomenon, and that these improvements have substantial economic value.

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1. Introduction

Recent years have seen a substantial and growing interest in prediction markets as instruments for improving forecasts by appropriately aggregating and weighting information that is spread across many individuals. Prediction markets are organised so as to allow participants to trade contracts for which the payoffs depend on a given uncertain event, and the market prices can be interpreted as forecasts of the probability of the event (Paton, Siegel, & Vaughan Williams, 2009). The ability of prediction market prices to reflect all relevant information fully is traditionally grounded on the efficient market hypothesis (EMH) (Fama, 1970). This assumes that decision-makers in financial markets assess the likelihoods of all possible future outcomes rationally and make financial asset allocations that represent their degree of beliefs optimally, having taken into consideration their risk–reward trade-offs. It has been argued that prediction markets are among the most efficient mechanisms for aggregating asymmetrically-dispersed private and public information, as decision-makers in these markets have incentives to continue trading until the information that they hold is incorporated fully into market prices (Spann & Skiera, 2009). This argument is supported by the growing body of evidence that the probabilities derived from prediction markets outperform sophisticated forecasting methods (Spann, 2003; Tziralis & Tatsiopoulos, 2007), and by the fact that increasing numbers of corporations use prediction markets as decision support tools (Cowgill, Wolfers, & Zitzewitz, 2009; Soukhoroukova, Spann, & Skiera, 2012).

Despite the clear strengths of prediction markets, they have been shown to suffer from pricing anomalies, where final prices can fail to reflect all relevant information appropriately. These anomalies include the favourite-longshot bias (FLB) (where decision-makers systematically under-/overestimate the winning probabilities of the most/least likely outcomes) (Cain, Law, & Peel, 2002; Smith & Vaughan Williams, 2010), and pricing anomalies that arise from herding (Soosung & Salmon, 2004), over- and under-reactions to new information (Poteshman, 2001),

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anchoring (Johnson, Schnytzer, & Liu, 2009), and representativeness bias (Tassoni, 1996).

The psychology and decision-making literatures suggest that such pricing anomalies can often arise from participants' moods, emotions and feelings[1] (referred to collectively hereafter as mood). Empirical research has shown that 'mood misattribution' can occur, whereby current, transient factors that are not related to the decision can affect mood, and this can lead to judgments that depart from those expected from fully rational decision makers (Lucey & Dowling, 2005). Furthermore, it has been shown that mood can be influenced by weather and atmospheric conditions (referred to hereafter as environmental conditions (EC)). As a consequence, mood fluctuations caused by EC can potentially decrease the ability of decision-makers to make probabilistic judgments that account for all relevant rational considerations, and can diminish their ability to effectively learn from feedback. Mood fluctuations caused by EC can potentially impair these key ingredients of effective prediction markets (Vosgerau, 2010), and may therefore reduce the accuracy of forecasts derived from these markets.

Individuals are often unaware of the extent to which EC affect their moods, and underestimate the degree to which these in turn affect the quality of their decisions, especially when facing complexity, risk and/or uncertainty (Bechara, Damasio, Tranel, & Damasio, 1997; Loewenstein, 2000). As a consequence, the degree to which EC affect the accuracy of prediction market forecasts may have been neglected. We help shed light on this aspect of prediction markets by examining both the extent to which EC systematically affect the quality of forecasts derived from prediction markets and the extent to which it is possible to improve forecasts derived from these markets by accounting for EC.

We explore whether or not EC can affect the quality of predictions derived from prediction markets by choosing a market in which these factors have a very good chance of not being influential. If EC affect the quality of predictions in this market, then such is likely also to be the case in a wide variety of other prediction markets. Specifically, we examine a prediction market that is renowned for the accuracy of its forecasts, namely the horserace betting market. In fact, the forecasts from sports betting markets have been shown to outperform expert predictions (Forrest & Simmons, 2000), statistical models using fundamental variables (Benter, 1994), and aggregated fast and frugal predictions made by laypeople (Serwe & Frings, 2006; Spann & Skiera, 2009). The accuracy of the forecasts derived from these markets has been tied to the fact that participants can engage in large numbers of similar markets (Paton & Vaughan Williams, 2005), which enables them to learn the factors that influence horseracing performances. An additional benefit that these markets offer for examining our research question is that they have a specific contract end-point, at which time all uncertainty is resolved (an unequivocal outcome occurs), thus enabling a clear assessment of the forecast accuracy.

In summary, these prediction market conditions offer an ideal setting in which to examine our research questions, one in which the prediction market participants have a stronger likelihood of not being influenced by EC than in many other market settings. Therefore, if we find that environmental conditions affect the accuracy of predictions derived from prices in horserace betting markets, we can be fairly confident that such will also be the case in other prediction markets. Furthermore, the unequivocal nature of the outcome in these markets allows us to test the extent to which adjusting prices to account for environmental factors leads to better forecasts.

Our results demonstrate that the accuracy of the probability forecasts derived from final prices in horserace betting markets can be improved substantially by understanding and correcting for situations in which we expect EC to affect prices. Specifically, our results reveal that: (i) pricing anomalies associated with EC are present in prediction markets, since the accuracy of the forecasts derived from final prices appears to be affected by mood misattribution; (ii) the forecasting accuracy can be improved significantly by correcting for this mood misattribution; and (iii) recognising and correcting probability forecasts for mood misattribution can lead to substantial economic gains.

The remainder of the paper is organised as follows. Section 2 discusses the influence that EC may have on decision making, and we use this discussion to motivate our hypotheses. Section 3 discusses the features of the different mechanisms of sports prediction markets, and introduces the data used in this research. Section 4 describes the methods employed for testing our proposed hypotheses. The empirical results are reported and discussed in Section 5. Finally, Section 6 draws conclusions and identifies important implications of our research both for prediction markets and for wider decision-making contexts.

2. Environmental conditions, mood and prediction market forecasts

2.1. The role of mood, emotion and feelings in decision-making

The dual-process theory of decision-making and information processing suggests that there are two fundamental systems of thinking that operate in parallel and depend on each other for guidance when making decisions. In the analytic system, logic and normative rules prevail. This system normally takes effort and requires conscious control during the judgment process. In contrast, intuition, mood, emotions and feelings are thought to drive the experiential system of decision-making. This system involves quick information processing, since, for the most part, it is performed automatically by the subconscious mind. Mood, emotions and feelings are often the first reactions when processing information (because of the relatively higher process speed of the experiential system), and as a consequence, they subsequently provide initial guidance to the

[1] The distinctions between mood, emotion and feelings are not consistent in the psychology literature (Oatley & Jenkins, 1996). 'Mood' is defined as a distinctive emotional tone or attitude expressed for a short period of time. 'Emotion' is an affective state of mind, deriving from feelings, moods, sensations and relationships with others. 'Feelings' is a term that is often used to describe either mood or emotions. Thus, rather than attempting to distinguish the effects of mood, emotions and feelings, we simply refer to them collectively as 'mood' and examine their combined effects throughout the paper.

analytic system when assessing information and making cognitive evaluations regarding future outcomes (Slovic, Finucane, Peters, & MacGregor, 2004). In fact, the dual process theory postulates that fully rational analytic reasoning, as required by the EMH, cannot function effectively unless it is guided by mood (Kahneman & Frederick, 2002; Sloman, 1996; Zajonc, 1980).

Loewenstein, Weber, Hsee, and Welch's (2001) risk-as-feelings model supplements the dual-process theory by incorporating the idea that mood influences 'every' aspect of the decision-making process. This model incorporates the view that decisions under conditions of risk and uncertainty are evaluated largely at the cognitive level, based fundamentally on logical and rational outcome predictions and a cost–benefit analysis. However, the model postulates that mood, triggered by the anticipation of future outcomes, the evaluation of subjective probabilities and the environmental circumstances may exert an external influence on these evaluations.

In summary, the research suggests that mood is an important factor that affects decision-making. It can aid decision-making effectively, as is illustrated by the fundamental role played by intuition and instinct in enabling human survival and evolution. However, mood can also have a direct effect on the decision quality. In particular, the consensus that emerges from previous research is that a good mood leads to more optimistic judgments of future outcomes (Isen, Shalker, Clark, & Karp, 1978), a greater use of irrelevant information (Sinclair & Mark, 1995), a heavier reliance on the experiential system of thinking and on previous experiences, the use of more simplistic stereotyping and simplification heuristics (Forgas, 1995), less engagement in critical modes of thinking, and a greater susceptibility to distractions (Hirshleifer & Shumway, 2003). In contrast, decision makers who are in bad mood states tend to make more pessimistic judgments of future outcomes (Isen et al., 1978), undertake more critical information processing (Isen et al., 1978), engage in more analytical and reasoning activities, and react more efficiently to relevant news (Sinclair & Mark, 1995).

2.2. The effect of environmental factors on emotions, mood and feelings

The empirical evidence suggests that a range of environmental factors influence mood. The conventional view is that 'good' EC induce positive mood and 'bad' EC induce low mood (Lockard, McDonald, Clifford, & Martinez, 1976; Schwarz & Clore, 1983). For instance, higher temperatures and atmospheric pressures, clear skies, an absence of rain and geomagnetic storms, good air quality, and lower humidity and wind have all been linked to individuals experiencing positive mood states, whereas lower temperatures and atmospheric pressures, cloudy and rainy days, geomagnetic storms, poor air quality, higher humidity and wind have all been associated with low mood.[2]

There is also evidence that mood can be influenced by moon cycles. For instance, the higher illuminance levels emitted on full moon nights have been found to disrupt sleeping patterns, which in turn leads to negative moods (Armitage, 2007; Cajochen et al., 2013; Kelley, 1942). Furthermore, the shortening of daylight hours during the winter months has been linked to seasonal affective disorder (SAD), a medical condition which causes individuals to experience consistent low moods and depression (Kamstra, Kramer, & Levi, 2003).

2.3. Empirical evidence linking environmental conditions to market outcomes

Most studies that have investigated the extent to which EC affect decision making in a naturalistic environment, via their influence on mood, have been conducted in financial markets. However, studies that have explored the effect of the EC on equity returns[3] have reached no consensus regarding the direction of such influence. For instance, some studies have suggested that negative moods (e.g. during geomagnetic storms, cloudy and rainy days, after a full moon night, and during the months of autumn and winter) lead to more pessimistic judgments about future outcomes, causing investors to be more prone to selling stocks, and thus driving stock prices down and leading to negative returns. Some studies have also shown that positive moods can lead to more optimistic judgments about future outcomes, increasing investors' willingness to buy stocks, and thus driving stock prices up and leading to positive returns (Denissen, Penke, Butalid, & Aken, 2008; Goetzmann, Kim, Kumar, & Wang, 2015; Kamstra et al., 2003). However, other studies have suggested that negative moods are associated with positive equity returns, thus suggesting the opposite relationship between mood and returns.

Dowling and Lucey (2005) argue that although EC have an influence on optimism, they may have an even higher influence on the level of analytical reasoning adopted when making decisions. They argue that this is the key factor that causes good mood states to be associated with negative equity returns. For example, they suggest that investors in negative (positive) mood states are more likely to undertake more (less) critical information processing and to be more (less) likely to incorporate relevant information effectively when making their investment decisions – factors which lead to higher (lower) equity returns.

Hirshleifer and Shumway's (2003) study further illustrates the mixed conclusions concerning the relationship between mood and returns in financial markets. In particular, they provided evidence that temperature, rain and sunshine were correlated significantly with stock returns across 25 countries. However, the direction of their influence on returns varied depending on the location of the stock market studied.

In summary, several studies have attempted to explore whether mood misattribution affects stock returns. However, no definitive conclusions of such an effect can be drawn from these studies, as some researchers have found that EC appear to affect asset prices (e.g., Cao & Wei, 2005;

[2] For a detailed review of the mechanisms by which these occur, see Keller et al. (2005) on temperature and atmospheric pressure; Eagles (1994) on cloud cover; Bagby, Shuller, Levitt, Joffe, and Harkness (1996) on rain; and Persinger and Levesque (1983) on geomagnetic storms and wind speed.

[3] These have examined most EC, other than air quality.

Chang, Chen, Chou, & Lin, 2008; Hirshleifer & Shumway, 2003; Lucey & Dowling, 2005), whereas other studies indicate that EC have no such influence on asset prices (e.g., Goetzmann & Zhu, 2005; Jacobsen & Marquering, 2008; Pardo & Valor, 2003).

Shortcomings in previous research may have led to these mixed conclusions. In particular, there have often been large time differences between the trades taking place and the weather observations, most studies have not incorporated a wide set of EC variables, and there are difficulties in attempting to develop an unequivocal measure of the influence on market prices due to the infinitely lived nature of the assets being studied (e.g., stocks). In addition, no study in the context of financial markets has examined the influence of EC-induced mood misattribution on the calibration of prices.

Our research question is designed to fill this research gap, and we develop a methodology that is designed to overcome the shortcomings of previous research. This methodology is used to test the following two hypotheses, derived directly from the literature discussed above, which suggests that EC can influence mood and that this can lead to pricing anomalies, via mood misattribution:

H1. Environmental conditions that are expected to lead to good (bad) mood have a systematic, negative (positive) influence on the accuracy of forecasts derived from prices in prediction markets.

As is discussed in detail in Section 3.1, final contract prices, and consequently returns, in horseracing markets reflect the abilities of decision makers to predict the winning probabilities of horses accurately. Therefore, based on the literature presented here on the influence of EC on decision-making, we postulate that EC may influence the analytical effort displayed by decision-makers, which in turn may influence the calibration of their forecasts.

If we find evidence to support H1, we propose the following:

H2. The accuracy of probability estimates derived from final market prices in prediction markets can be improved by correcting for the influences of environmental conditions.

3. Data

3.1. Different prediction market mechanisms in sports markets

The mechanisms underlying prediction markets can vary. However, their ultimate purpose is to provide an appropriate means of aggregating individuals' beliefs as to the likelihood of future events (Vaughan Williams, 2011).

Previous sports prediction market research has largely examined pari-mutuel and betting exchange mechanisms. In the former, all bets on a particular event are combined together in a pool, after which the market organiser removes its commission and the payoffs are calculated so as to distribute the remaining betting volume among the winning bets. In this form of prediction market, the winning probability of each contestant is represented by the proportional dollar amounts placed by individuals on particular contestants. Betting exchanges, on the other hand, work in a similar manner to traditional financial markets, where individuals trade contracts between themselves, by either buying or selling a contract on a specified event. In this form of prediction market, the market organiser commission is charged only on winning contracts. Both of these forms of prediction market may be described as 'person-to-person' betting, in that the market organisers simply charge a commission to provide the infrastructure to allow individuals to trade against each other on the basis of their beliefs concerning the outcomes of events.

A relatively less studied prediction market mechanism is that of bookmaker markets, sometimes referred as 'quote-driven' prediction markets. In this market setting, bookmakers act as market makers, quoting the contract prices (odds) that they are willing to offer individuals to place bets on particular events or contestants, and individuals can either bet or not at the quoted prices. Thus, in addition to bettors, bookmakers are also important decision-makers in determining final contract prices, as they are financially susceptible to the outcome of the event (i.e., they are financially dependent on the outcomes of events, as they participate by taking the opposite side of every contract (bet) traded). The quoted prices are determined by the price-setting behaviour of the market-maker. For example, bookmakers might quote prices that aim to reflect the true outcome probabilities in order to avoid substantial losses, thus allowing them to earn a long-term profit that is equal to the average commission charged. The bookmakers can determine these probabilities based on their own assessments of the relevant information, together with the relative volumes of betting on different outcomes. Alternatively, bookmakers can quote odds that attract higher betting volumes on each outcome, such that they earn a profit equal to the commission charged, whatever the event outcome. These price-setting behaviours mean that a contract price in this prediction market setting reflects both the bookmaker's and the bettors' beliefs concerning the outcome of the event (Franck, Verbeek, & Nuesch, 2010).

The few studies that have compared the forecasting accuracies of different prediction market mechanisms have mainly compared the accuracies of bookmakers' odds and prices from the leading UK betting exchange, Betfair. The results have suggested that Betfair prices have a slightly greater predictive power (Franck et al., 2010; Strumbelj, 2014), but this difference is not statistically significant (Strumbelj, 2014). In fact, it is documented widely that quote-driven prediction markets are very efficient at aggregating individuals' beliefs and knowledge concerning future outcomes, as is evidenced by the good forecasting performance derived from final prices (Boulier & Stekler, 2003; Forrest, Goddard, & Simmons, 2005; Sung & Johnson, 2007). As a consequence, we employ bookmaker odds when testing our hypotheses.

3.2. Sources of data

The horseracing data were obtained from Raceform Ltd and cover all flat races in the United Kingdom between 2002 and 2016, inclusive. They consist of starting times, finishing positions, race classes, numbers of bends, an indicator for handicap (0 for non-handicap and 1 for handicap)

races, and bookmakers' starting prices (SP) for each of the 73,457 horses and 2717 jockeys in the 87,402 flat races run at 43 race tracks across the United Kingdom during this 15-year period. Races occur in all months of the year under different going conditions, but the majority of races take place between May and September (61.31%) and on good/fast conditions (79.9%).

We avoid overfitting and estimate accuracy improvements that may be possible from incorporating EC into forecasts by dividing the research sample into two parts: the in-sample (training set) data, which consists of races run between 2002 and 2013 and represents approximately 80% of the data, and the out-of-sample data (holdout sample), which consists of races run between 2014 and 2016.[4]

The EC data were obtained from the Met Office Integrated Data Archive System. The database contained hourly data from weather stations across the United Kingdom. The closest weather station to each individual racetrack was identified using their stations' respective zip codes, and these were used to identify the EC prior to each race start time. The EC were captured in variables measuring the temperature, wind speed, cloud cover, geomagnetic activity, humidity, atmospheric pressure, rain amount, air quality, and cycle of the moon. Our review of the psychological and medical literature revealed that geomagnetic storms and full moon days may lead to a deterioration in mood, thus improving the quality of decisions. As a consequence, following Dowling and Lucey (2005), we coded '*geomagnetic storms*' as either 1 or 0, based on whether geomagnetic activity is greater than or less than or equal to 29, respectively, with 1 indicating the occurrence of a geomagnetic storm. In addition, '*full moon*' was coded either 1 or 0 to indicate days on which a full moon does or does not occur, respectively. We also derived a numerical estimate of the seasonal affective disorder (*SAD*) based on Kamstra et al.'s (2003)[5] methodology. The descriptive statistics of the horseracing and EC variables are presented in Table 1.

4. Method

The testing of our hypotheses required us to investigate: (i) which EC factors influence the accuracy of forecasts derived from market prices, and (ii) the extent to which the forecasting accuracy derived from market prices can be improved by correcting for the influences of EC. We achieve this by first employing regression analysis to determine the EC factors that influence the Brier scores (a widely adopted score function that measures the accuracy of probabilistic forecasts), which enabled us to test H1. We test H2 by employing conditional logit models for measuring the degree to which the forecast accuracy can be improved by correcting for the probable influences of EC on probability estimates derived from market prices. This is achieved by comparing the accuracy of probability forecasts generated by conditional logit models that incorporate the following predictors: (a) EC and probabilities derived from market prices and (b) probabilities derived from market prices. The coefficients of these models, estimated on the basis of the training sample data, are used to forecast winning probabilities for the holdout sample. We then test H2 by basing Kelly betting strategies on the probability estimates derived from these models and comparing the returns obtained. We now examine the details of our methodology further.

4.1. Deseasonalized variables

Acclimatization is an important biological mechanism that needs to be considered when studying decision-making, as it moderates the influence of EC on mood. In particular, there is evidence that individuals have the ability to acclimatize to seasonal changes in the environment, which can potentially either reduce or exacerbate the expected influence of EC on mood (Young, Muza, Sawka, Gonzalez, & Pandolf, 1986). Furthermore, acclimatization also allows individuals to maintain the stability of internal functions across a range of different EC (Hancock & Vasmatzidis, 2003). Consequently, the influence of the raw EC on decision-making may be moderated by an individual's ability to acclimatize to the current seasonal conditions. This can potentially lead to raw EC having a different influence on mood at different times of the year (e.g. an above-average temperature may improve mood during the winter but cause a deterioration in mood in the summer). We address this by following Hirshleifer and Shumway (2003) methodology and deseasonalizing[6] the EC that are included in the regression analyses (with the exception of geomagnetic storms, full moon and SAD, as they do not follow seasonal patterns). The expectation is that the deseasonalized variables will provide a better measure of the influence of EC on mood (and consequently on decision-making), as EC are adjusted to represent better the ability of individuals to acclimatize.

In addition, the process of deseasonalyzing EC variables reduces the seasonal correlation among EC variables. For example, deseasonalyzing the weather variables ensures that deteriorating weather conditions would not be observed exclusively during winter months, and *vice versa* for summer months. In particular, deseasonalyzing an EC variable means that deteriorating weather conditions should

[4] We ensured the robustness of our analysis to different cut-off points when determining the training and holdout samples by conducting our analysis on different training and holdout samples, consisting of different racing calendar years, and compared the results with those reported in Tables 3–5. The summary statistics (i.e., adjusted and pseudo R^2, signs of coefficients and their respective significance levels) remained consistent with the results reported in Tables 3 and 4. Equally, the relative economic performances of the betting strategies presented in Table 5 remained consistent across different samples. The results of these robustness checks suggest that the results presented in Tables 3–5 are robust to the use of different cut-off points for dividing the sample into training and holdout datasets.

[5] Derived using $SAD_t = \begin{cases} \left\{24 - 7.72\, x\, arcos\left[-\tan\left(\frac{2\pi\sigma}{360}\right) \times \tan\left(0.4102\, x\, sin\left(\left(\frac{2\pi}{365}\right) \times (Julian_t - 80.25)\right)\right)\right]\right\} - 12 \text{ for racing days in the fall and winter} \\ Zero\ Other\ wise, \end{cases}$

where Julian ranges from 1 to 365(6), representing the number of the day in each year, and δ is the latitude (in degrees) of race tracks.

[6] The deseasonalization procedure involves subtracting the monthly average from the raw environmental condition in question.

Table 1
Descriptive statistics of horseracing and environmental condition variables.

Variable	Mean	Std. Dev.	Min	Max
Bookmaker commission (%)	0.17	0.08	0.01	0.95
No. runners	10.01	3.56	2	36
Race class	4.59	1.39	1	7
No. bends	1.52	1.53	0	8
Temperature (° C)	14.25	5.68	−4.40	33.20
Wind speed (Km/h)	9.04	4.58	0	36
Cloud (varies from '0' clear sky to '9' complete cover)	4.98	3.04	0	9
Humidity (%)	70.54	16.40	19.30	100
Atmospheric pressure (hPa)	1014.62	9.70	965.80	1044.30
Rain amount (mm)	0.10	0.60	0	28
Air quality (varies from '1' for perfect air quality to '10' for hazardous air quality)	3.15	0.95	1	10
Geomagnetic storms ('1' indicates days of geomagnetic storms, '0' otherwise)	0.42	0.49	0	1
Full moon ('1' indicates days after a full moon night, '0' otherwise)	0.23	0.42	0	1
SAD (indicates the lengthening of night hours, from the expected 12 h, for different days of the year and geographic locations)	0.80	1.42	0	5.05

occur if the EC variable in question is below the *expected* weather condition for a particular month. Thus, deteriorating conditions can be observed all year round, not only during winter months. This process of deseasonalyzing weather variables has also been used by several authors for reducing the correlations among weather variables (for an example, see Lu & Chou, 2012).

4.2. Combining environmental conditions variables

A statistical model that includes a wide range of EC may lead to spurious and biased estimates due to underlying associations among the variables (Jacobsen & Marquering, 2008). Temperature, wind, atmospheric pressure, cloud cover, humidity, rain and air quality, for example, are not independent, as changes in any one of these variables may affect the others (Ahrens, Jackson, & Jackson, 2012). In particular, a reduction in atmospheric pressure can lead to rain, and rain and cloud cover are highly correlated. One way of handling this issue would be to discard EC variables that have strong associations. However, our review of the psychology and decision-making literatures revealed that mood can be influenced by a combination of different EC (e.g. Denissen et al., 2008; Persinger & Levesque, 1983; Tarquini, Perfetto, & Tarquini, 1998;).

In these circumstances, discarding EC variables may lead to a model that fails to represent the true underlying influence of EC on prediction market prices. An alternative approach is to adopt a data reduction technique. We believe that this approach is more suitable for our study, as all variables are retained in data reduction. We achieve this by employing principal component analysis, which allows the possibility of retaining information from all EC variables that are observed in a particular race. Principal component analysis aggregates the interdependent EC variables into a smaller set of uncorrelated composite variables, called principal components (PC). It achieves this by defining an orthogonal linear weighted combination of the EC variables that are observed on race j:

$$\begin{aligned} PC_1 &= \alpha_{11}X_{1j} + \alpha_{12}X_{2j} + \cdots + \alpha_{1n}X_{nj} \\ &\vdots \\ PC_m &= \alpha_{m1}X_{1j} + \alpha_{m2}X_{2j} + \cdots + \alpha_{mn}X_{nj}, \end{aligned} \tag{1}$$

where α_{mn} is the weight for the mth principal component and the nth EC variable X. The weights for each PC are defined by the eigenvector of the correlation matrix of the EC variables. The PCs are defined in such a way that the first component accounts for the largest amount of the variance σ^2 among the EC variables, where the sum of the squared weights $\alpha_{11}^2 + \alpha_{12}^2 + \cdots + \alpha_{1n}^2$ for each PC is equal to one. The proportional variance that is accounted for by each PC is given by σ_m^2/n. PCs with resulting eigenvalues that are lower than one account for less of the variance among the EC variables than was contributed by an individual EC variable. This may render the components' scores unreliable (Kaiser, 1958). As a consequence, we adopt the Kaiser criterion and consider only PCs that achieve eigenvalues greater than one.

After establishing which components are to be retained, a varimax orthogonal rotation with Kaiser normalization is performed in order to constrain the PCs to be uncorrelated. The PC analysis reveals that three components achieved eigenvalues greater than one. These three components, as shown in Table 2, account for 61.63% of the total variance among the deseasonalized EC. We label the resulting PCs in a manner that best represents the highest component weights. As a consequence, component 1 is labelled '*warmer, sunnier and drier conditions*' (based on component weights of 0.533, −0.447 and −0.534 for the deseasonalized *temperature*, *cloud cover* and *humidity*, respectively), component 2 is labelled '*deteriorating weather*'[7] (based on component weights of 0.767 and –0.62 for the deseasonalized *wind speed* and *air pressure*, respectively), and component 3 was labelled '*wetter weather and poorer air quality*' (based on component weights of 0.696 and 0.548 for the deseasonalized *rain* and *air quality*, respectively). The resulting three components were used as predictors in the subsequent analyses, in addition to *SAD*, *geomagnetic storms* and *full moon*.

[7] This label reflects the understanding that a lower atmospheric pressure leads to higher wind speeds, and higher wind speeds are associated with deteriorating weather conditions, such as cloudier and rainier days (Trujillo & Thurman, 2001).

Table 2
Principal component factor loadings with Kaiser normalised varimax rotation based on deseasonalized environmental conditions.

Environmental condition variable (deseasonalized)	Component 1 Warmer, sunnier and drier weather	Component 2 Deteriorating weather	Component 3 Wetter weather and poorer air quality
Temperature	0.533	0.063	0.189
Air quality	0.419	−0.073	0.548
Wind speed	0.115	0.767	−0.210
Pressure	0.124	−0.620	−0.254
Humidity	−0.534	−0.072	0.262
Cloud cover	−0.447	0.092	0.045
Rain	−0.165	0.067	0.696
Eigenvalue	1.99	1.28	1.05
Proportional variance explained	0.2838	0.1820	0.1505
Cumulative variance explained	0.2838	0.4658	0.6163

4.3. Preference variable

A horse's performance is determined by a large range of factors (Benter, Miel, & Turnburgh, 1996). Based on the medical and psychological literature, Costa Sperb, Sung, Ma, and Johnson (2017) predicted that EC are likely to impact the performances of both horses and jockeys, due to their influence on the horse's and jockey's physiological and metabolic capabilities and on their current state of mind/mood. As a result, the authors developed a methodology that forecasts a horse's winning probability based on a range of EC. They refer to this as a 'preference variable' methodology, since it identifies the EC under which horses and jockeys perform well (i.e., for which they display a 'preference'). As a consequence, when attempting to capture the influence that EC may exert on the final prices in prediction markets, it is necessary first to control for the possible effects of these conditions on horses' and jockeys' performances. Including these 'preference variable' in our model allows us to control for the influence of EC on the performance of horses and jockeys. Thus, this approach enables us to isolate the influence of EC on the calibration of final market prices (potentially resulting from the effects of the EC on the bettors' moods).

4.4. Assessing influence of environmental conditions on forecast accuracy

We test H1, namely that EC have a systematic influence on the accuracy of forecasts derived from prices in prediction markets, by employing the Brier score (Brier, 1950) as our measure of the forecast accuracy. This is selected because it is used widely to assess the accuracy of probabilistic, mutually exclusive discrete outcomes in sports (e.g. Corral & Rodriguez, 2010; McHale & Morton, 2011; Strumbelj & Sikonja, 2010).

The Brier score for a total of N runners in a race is defined as

$$BS_j = \frac{1}{N}\sum_{i=1}^{N}\left(f_{ij} - o_{ij}\right)^2, \tag{2}$$

where f_{ij} is the odds implied probability[8] for horse and jockey pair i in race j and o_{ij} is the actual outcome of the race (i.e., 1 for the winner and 0 otherwise). Brier scores can vary between 0 and 1, where 0 represents a perfect prediction accuracy (i.e., lower Brier scores indicate better forecasts). Importantly, the Brier score is minimized when the true probabilities are estimated, and therefore provides a robust measure of the prediction accuracy derived from final prices.

We establish whether EC influence the quality of the probability estimates derived from the final prices by determining the extent to which the Brier score is influenced by three key variables: *Horseracing Factors*, *Preference Variable Variance* and *Environmental Conditions*. This is achieved by using the training data to estimate the following model:

$$\begin{aligned}\text{Brier score}_j = {} & \alpha + \beta_1 \textit{Preference Variable Variance}_j \\ & + \beta_k \textit{Horseracing Factors}_j + \\ & \beta_k \textit{Environmental Conditons}_j + \varepsilon.\end{aligned} \tag{3}$$

We now define each of the three variables used in this model.

Horseracing Factors: Previous research has indicated that the accuracy of the forecasts derived from prices in horseracing markets may be influenced by race-specific factors such as the number of runners (Gramm & Owens, 2005), whether the race is a handicap (Brown, 2016), the number of bends that horses encounter in the race (Johnson, O'Brien, & Sung, 2010), the class of the race (Sung, Johnson, & Peirson, 2012), whether the race is run on a weekend (Sung et al., 2012), whether the race is the first or last of the day at a particular racetrack (Johnson & Bruce, 1993), and the bookmaker commission charged on the race (Vaughan Williams & Paton, 1998). We include all of these factors in the regression as control variables, referred to collectively as 'Horseracing Factors'.

Preference Variable Variance: The accuracy of the forecasts derived from final prices in betting markets is sensitive to the level of uncertainty regarding the race outcomes (Moul & Keller, 2014). We believe that this uncertainty is likely to be linked to the variance of the preference variable

[8] The odds implied probabilities f are the probabilities derived from final betting prices (SP), where the probabilities for a particular race are constrained to sum to 1, defined as $f_{ij} = \frac{\left(\frac{1}{SP_{ij}+1}\right)}{\left(\sum_{i=1}^{N}\frac{1}{SP_{ij}+1}\right)}$.

probabilities across horses in a given race. For example, a lower variance observed for the horses' preference variable probability estimates in a given race suggests that it is more difficult to distinguish the influence of EC on the performances of individual horses and jockeys (i.e., a higher outcome uncertainty). We believe that this is likely to lead to poorer, less calibrated judgments, which in turn may suggest that the accuracy of forecasts derived from the market prices will be lower. As a consequence, we control for the influence of preference variable variance on the forecasting accuracy by including the *Preference Variable Variance* in our regression (Eq. (3)).

Environmental Conditions: We assess the impact of EC on the forecast accuracy by including a range of EC variables in the regression (Eq. (3)). In particular, we include the PCA components that are derived in the manner outlined in Section 4.2, together with variables for capturing SAD, geomagnetic storms and full moons. A statistically significant negative coefficient for the EC variables associated with poorer mood (namely the *wetter weather and poorer air quality* and *deteriorating weather* components, geomagnetic storms, full moon and SAD), and a statistically significant positive coefficient for the EC variable associated with better mood (i.e. *warmer, sunnier and drier weather* component) would be indicative of a mood misattribution bias, thus providing evidence in support of H1.

4.5. Improving the forecasting accuracy by accounting for the influence of environmental condition variables on market prices

We test H2 by examining the extent to which the accuracy of forecasts derived from market prices can be improved by correcting for a mood misattribution bias. We achieve this by employing the modelling procedure that is used most widely for assessing the degree to which odds in horseracing prediction markets reflect all available information, namely the conditional logit (Johnson & Bruce, 2001). The aim of the conditional logit model (referred to as CL hereafter) is to derive the winning probability p_{ij} of a horse and jockey pair i in race j such that the sum of the winning probabilities for all horses in each race is constrained to be one. These probabilities are estimated using a vector of h predictors $\mathbf{Y}_{ij} = \left[Y_{ij}^1, \ldots, Y_{ij}^h\right]$, which capture information with respect to each jockey and horse pair i in race j. One particular advantage of this model is that the winning probability estimates for each horse are conditional on the competitiveness of the race.

We estimate the CL model by deriving a 'winningness' index W_{ij} for every horse and jockey pair i in race j, such that

$$W_{ij} = \sum_{r=1}^{h} \beta_r Y_{ij}^r + \varepsilon_{ij}, \tag{4}$$

where β_r is a coefficient that measures the importance of predictor $Y_{ij}{}^r$ in determining the likelihood of horse and jockey pair i winning race j, and ε_{ij} is an independent error term that is distributed according to the double exponential distribution. W_{ij} is calculated such that the horse and jockey pair that is forecast to win a particular race is determined to be the one with the highest winningness index in that race. Thus, the estimated probability of the horse and jockey pair K winning race j (p_{Kj}), composed of N_j runners, is estimated by:

$$p_{Kj} = Prob(W_{Kj} > W_{ij}, i - 1, 2 \ldots N_j, i \neq K), \tag{5}$$

and therefore

$$p_{Kj} = Prob\left(\sum_{r=1}^{h} \beta_r \left(Y_{Kj}^r\right) + \varepsilon_{Kj} > \sum_{r=1}^{h} \beta_r \left(Y_{ij}^r\right) + \varepsilon_{ij}, \right.$$
$$\left. i = 1, 2 \ldots N_j, i \neq K \right). \tag{6}$$

The W_{ij} cannot be observed directly. However, whether horse i will win race j can be observed as a win/lose binary variable t_{ij} that is defined such that:

$$\begin{cases} t_{ij} = 1 \; if \; W_{ij} = Max\left(W_{1j}, W_{2j,} \ldots, W_{N_j j}\right) \\ t_{ij} = 0 \; otherwise. \end{cases} \tag{7}$$

The probability of the horse and jockey pair K winning race j can be represented by:

$$p_{Kj} = Prob\left(t_{Kj} = 1 | \left(p_{ij}\right), i = 1, 2 \ldots, N_j\right), \tag{8}$$

such that the conditional winning probability for the horse and jockey pair i in race j can be derived as

$$p_{ij} = \frac{exp\left(\sum_{r=1}^{h} \beta_r Y_{ij}^r\right)}{\sum_{i=1}^{N_j} exp\left(\sum_{r=1}^{h} \beta_r Y_{ij}^r\right)}, \tag{9}$$

where the β_r are estimated using maximum likelihood procedures.

We test H2, namely that the accuracy of the probability estimates derived from the final market prices in prediction markets can be improved by correcting for the likely influences of EC, by estimating three separate CL models using the training data. We then compare the abilities of these models to predict winning probabilities. The first, called 'benchmark CL', incorporates the winning probabilities derived from market prices as a single predictor. The second model, called 'preference CL', incorporates the winning probabilities derived from market prices and EC preference variable (as outlined in Section 4.3) as predictors. Lastly, the model called 'EC CL' incorporates as predictors the winning probabilities derived from market prices, the EC preference variable, and interaction terms between the EC principal components and market price probabilities.

A statistically significant coefficient for the EC preference variable in the EC CL model indicates that market prices do not incorporate information concerning the impact of EC on the performances of horses and jockeys fully. Statistically significant interaction terms between the EC principal components and market price probabilities provide supplementary evidence of a mood misattribution bias, as all relevant information concerning the impact of EC on the performances of horses and jockeys should have been discounted already by market prices and the EC preference variable.

We use coefficients estimated for these three CL models (i.e., 'benchmark CL', preference CL' and 'EC CL') based

on the training sample to develop winning probability estimates for the holdout sample races. These are then used as the basis of three separate Kelly betting strategies (Kelly, 1956) on the holdout sample races. Kelly betting is employed because its final performance depends directly on the accuracy of the forecast probabilities (MacLean, Ziemba, & Blazenko, 1992). Comparing the returns that can be achieved by the three Kelly betting strategies provides a means of examining the predictive value of correcting for any possible mood-induced misattribution bias (i.e., exploring the extent to which the returns from employing the EC CL model outperform those that can be achieved by employing the benchmark CL and preference CL models), thus providing a further test of H2.

Kelly betting assumes that some fraction $f_j(i)$ of wealth is bet on the horse and jockey pair i in race j. Let $f_j = \sum_{i=1}^{N_j} f_j(i)$ be the total fraction of wealth bet on race j with N_j runners. Given that the horse and jockey pair K wins race j (with odds O_{Kj}), the current wealth is projected to increase by a factor of $1 - \sum_{i=1}^{N_j} f_j(i) + f_j(K)(O_{Kj} + 1)$. Kelly betting involves selecting the value of f_j that maximises the expected log of winnings, $F(f_j)$, such as:

$$F\left(f_j\right) = \sum_{K=1}^{N_j} p_{Kj}^{\varphi} log\left(f_j(K)\left(O_{Kj}+1\right) + 1 - \sum_{i=1}^{N_j} f_j(i)\right), \quad (10)$$

where p_{Kj}^{φ} are the predicted winning probabilities (as estimated by the 'benchmark CL', 'preference CL' and 'EC CL' models). Thus, Kelly betting selects bets that maximize the expected log returns over all potential winners using the input model probabilities p_{Kj}^{φ}.

Employing the Kelly betting strategy can lead to very large bets being recommended as wealth levels increase later in the sequence of bets, or as a consequence of bets in which a large fraction of wealth ($f_j(i)$) is prescribed (Benter, 1994). We avoid the possibility of a betting strategy's success being biased artificially by the sequence of betting or the result of one or two large bets by employing a fractional Kelly strategy without any re-investment of winnings. Specifically, the bank size that is used to calculate the bet size is returned to its initial amount after each bet, independent of the outcome of that bet. In addition, we employ a 0.5 Kelly strategy, whereby we bet 50% of the recommended Kelly bet ($f_j(i)$) in a given race. This is done in order to prevent individual large bets from altering the final economic performance of the betting strategy substantially. As a consequence, adopting a 0.5 Kelly without reinvestment of the winnings ensures that the performance of a betting strategy is representative more of the collective forecast value of the predicted winning probabilities (p_{Kj}^{φ}) than of the (un)fortunate outcomes of a few large bets.

An initial wealth of $1000 is assumed for all three Kelly betting strategies, and we measure the success of each betting strategy by determining the total increase from the initial wealth as a result of applying the strategy. If a Kelly betting strategy on the holdout data produces better returns when it is based on winning probabilities estimated using the EC CL model (incorporating market prices, EC preference variable and environmental conditions) than when it is based on probability estimates from the 'benchmark CL' and 'preference CL' models, this provides further evidence in support of H2. Specifically, this implies that significantly higher returns can be achieved by correcting for the influence of EC on the quality of the probability estimates derived from market prices.

5. Results

The first set of results relates to our tests for detecting any possible EC-induced mood misattribution bias in the winning probability forecasts derived from the final market prices. The second set of results aims to examine the extent to which it is possible to improve the probability estimates derived from market prices by correcting for any misattribution bias that may be present.

5.1. The influence of environmental conditions on the accuracy of forecasts derived from prices in prediction markets

Table 3 summarizes the results of estimating a linear regression in the form of Eq. (3) in order to examine the impact of EC on the accuracy of forecasts derived from prices in prediction markets. These results show that six of the eight horserace factors have a significant effect on the forecast accuracy, as measured by the Brier score. This suggests that it is important to control for these factors when assessing the impact of EC on the accuracy of forecasts derived from prices in prediction markets.

The significant positive coefficient of the *preference variable variance* indicates that larger disparities between the effects of EC on the performances of individual horses and jockeys in a given race lead to market prices that deviate further from their correct value (i.e., the forecasts derived from these prices will be less accurate). This suggests that decision makers are not identifying the effects of EC on the performances of individual horses and jockeys appropriately, because greater disparities between the effects of EC on the performances of individual horses and jockeys should enable market participants to better distinguish individual horses'/jockeys' chances of success, thereby improving the forecasting accuracy.

The results relating to the influences of the six EC variables on the forecasting accuracy are revealing, and suggest that mood misattribution may affect the accuracy of forecasts derived from final prices, thus providing evidence in support of H1. Specifically, the *warmer, sunnier and drier weather* and *deteriorating weather* EC components and *SAD* are significant in explaining the forecasting accuracy (t-values of −2.15, −2.00 and −1.96, respectively), even after controlling for horserace factors and the *preference variable variance*.

The negative coefficients for the *deteriorating weather* component and *SAD* are consistent with the notion that poorer moods are associated with more analytical and logical reasoning. In particular, these results suggest that poorer moods, induced by the EC related to these variables, are associated with more accurate forecasts being derived from market prices (i.e., lower Brier scores).

However, the negative coefficient for the *warmer, sunnier and drier weather* component suggests the opposite. Specifically, one might expect that a positive mood induced by higher temperatures and lower cloud cover and humidity would lead to a greater reliance on the experiential

system of thinking, thus leading to a poorer forecast calibration (i.e., a positive coefficient, indicating larger Brier scores). However, we find the opposite effect. This unexpected result may arise because the *warmer, sunnier and drier weather* component is not allocating weights exclusively to the EC variables associated with better mood. For instance, this component has a large and positive weight on air quality, and larger values of the air quality are associated with poorer mood. This factor may moderate, or perhaps in our case even outweigh, the positive influence on mood of the other large EC component weights, leading to in the negative coefficient observed for this component. Importantly, this may demonstrate the relevance of the combined influences of different EC on mood, thus illustrating the importance of including all relevant EC variables when assessing their effects on decision-making.

Importantly, the fact that we find a systematic relationship between EC and the accuracy of forecasts derived from market prices, even after the impacts of these EC on the performances of horses and jockeys have been taken into account, suggests that it may be possible to improve the probability estimates derived from market prices by correcting for the influence of the identified bias.

5.2. Improving the accuracy of forecasts derived from market prices by accounting for the influence of environmental conditions

We evaluated the extent to which forecasts of the winning probabilities derived from market prices can be improved by correcting for any EC-induced mood misattribution bias. We achieved this by estimating three CL models using the training data and comparing their abilities to predict winning probabilities. The first CL model, labelled 'benchmark CL', was estimated using market prices as a single covariate. The second CL model, labelled 'preference CL', was estimated using market prices and preference variable as covariates. Lastly, a CL model labelled 'EC CL' was estimated using market prices, preference variable and interaction terms between EC variables and market prices as covariates. The results of estimating these three CL models are presented in Table 4.

The results of estimating the 'benchmark CL' model indicate that market prices (z-score of 196.75) alone are highly significant in explaining winning probabilities (pseudo-R^2 = 0.1627). This result is in line with the evidence that decision-makers in sports prediction markets are amongst the most sophisticated forecasters, and that the final prices from this particular market provide a good guide to winning probabilities (Figlewski, 1979; Smith & Vaughan Williams, 2010).

The results of estimating the 'preference CL' show that the coefficient for the market price probabilities remains highly significant (z-score of 193.52), indicating that again, market prices alone are highly significant predictors. The *EC preference variable* was also significant at the 1% level (z-score 2.62), indicating that market prices are not accounting for the influence of the environment on performance fully. The positive coefficient of this *EC preference variable* is what one might expect, suggesting that horses and jockeys with stronger preferences for the EC experienced on the day of the race are more likely to win.

In the 'EC CL' model, market prices and the *EC preference variable* are still highly significant (z-scores of 124.57 and 2.61, respectively). In addition, the *warmer, sunnier and drier weather* component and the *deteriorating weather* component were also significant at the 1% level (z-scores of 2.61 and 2.56, respectively). These results are consistent with the regression analysis presented in Table 3, and suggest that EC have an influence on the quality of the forecasts derived from final market prices in this prediction market. *Geomagnetic storms* and *SAD* are significant only at the 10% level (z-scores of 1.83 and 1.73, respectively).

Taken together, these results provide a further indication of a mood misattribution bias being present in this prediction market, as all relevant influences of EC on performance should be discounted fully in market prices and the EC preference variable. However, the results indicate that although market price probabilities are still highly significant, the *EC preference variable* incorporates fundamental performance-related information concerning the preferences of horses and jockeys for certain EC that is not contained in market prices. Furthermore, even after controlling for horses' and jockeys' preferences for certain EC, the EC still influence the predictive power of market prices systematically. In particular, the model that incorporates the interaction terms between market price probabilities and EC ('EC CL') explains the results of races in the training dataset better than the 'preference CL' and 'benchmark CL' models (pseudo-R^2 of 0.1629, 0.1628 and 0.1627 for the 'EC CL', 'preference variable CL' and 'benchmark CL' models, respectively). This suggests that the accuracy of the forecasts derived from market prices can be improved by correcting for the mood misattribution bias detected.

We confirmed this by conducting log-likelihood ratio tests (LR) to examine whether the difference in explanatory power that arises from incorporating the EC-related information is statistically significant (see Johnson et al., 2009). In particular, we calculated $LR_1 = 2[(L_P) - (L_B)]$, where L_P and L_B are the log-likelihoods of the 'preference CL' and 'benchmark CL' models. We found that $LR_1 = 8.86$ ($\chi^2{}_1[0.01] = 6.63$), suggesting that market prices are not incorporating the impact of EC on the performances of horses and jockeys fully. We also calculated $LR_2 = 2[(L_{EC}) - (L_P)]$, where L_{EC} and L_P are the log-likelihoods of the 'EC CL' and 'preference CL' models, and found that $LR_2 = 18.80$, ($\chi^2{}_6[0.01] = 16.81$). This suggests that market prices suffer from a mood misattribution bias. In particular, the same EC still have a significant effect on the winning probability forecasts derived from market prices, even after controlling for the influence of EC on performance.

Taken together, these results indicate that the forecasts of winning probabilities that are derived from market prices are highly predictive of final race outcomes. They also demonstrate that, whilst participants in this betting market largely take into account the influence of the EC on the performances of horses and jockeys, the quality of their decisions is affected by environmental conditions. In addition, they suggest that we should be able to improve the probability forecasts derived from market prices significantly by controlling for the influence of EC on performance, as well as by correcting prices for the presence of a

Table 3
The influence of environmental conditions on the Brier scores (i.e. forecast accuracy) of forecasts derived from market prices.

	Variables	Coef.	Std. error	*t*-value
Environmental conditions	*Intercept*	0.16216	0.00200	81.08**
	Warmer, sunnier and drier weather component	−0.00019	0.00009	−2.15*
	Deteriorating weather component	−0.00022	0.00011	−2.00*
	Wetter weather and poorer air quality component	−0.00003	0.00012	−0.27
	Geomagnetic storms	0.00027	0.00025	1.07
	Full moon	0.00019	0.00029	0.64
	SAD	−0.00018	0.00009	−1.96*
Horseracing factors	*Bookmaker commission*	−0.00419	0.00188	−2.23*
	No. runners	−0.00654	0.00004	−149.16**
	Handicap races	0.00728	0.00026	27.49**
	Weekend races	0.00016	0.00029	0.54
	Last race	0.00010	0.00037	0.26
	First race	−0.00104	0.00036	−2.91**
	Race class	−0.00168	0.00010	−17.47**
	No. Bends	−0.00039	0.00008	−4.73**
Preference	*Preference variable variance*	0.34539	0.04887	7.07**
	Adjusted R^2	0.3513		

Notes:
* Indicates significance at the 5% level.
** Indicates significance at the 1% level.
Negative (positive) coefficients indicate lower (higher) Brier scores, thus representing more (less) accurate forecasts. Tests for multicollinearity among the explanatory variables reveal variance inflation factors (VIF) and tolerances well within acceptable limits (max values of 1.64 and 0.62, respectively; see Blaikie, 2003.

Table 4
Results of estimating the benchmark CL, preference CL, and EC CL models based on the training data (1 Jan, 2002 –31 Dec, 2013).

	Benchmark CL			Preference CL			EC CL		
	Coefficient	Std. error	*Z*-score	Coefficient	Std. error	*Z*-score	Coefficient	Std. error	*Z*-score
Market price probabilities	1.15342	0.00586	196.75**	1.150782	0.005947	193.52	1.14072	0.00916	124.57**
EC preference variable				0.062071	0.023723	2.62	0.06184	0.02371	2.61**
Market price probs. × *warmer, sunnier and drier weather* component							0.01093	0.00419	2.61**
Market price probs. × *deteriorating weather component*							0.01343	0.00525	2.56**
Market price probs. × *wetter weather and poorer air quality* component							−0.00457	0.00573	−0.80
Market price probs. × *geomagnetic storms*							0.02189	0.01199	1.83
Market price probs. × *full moon*							−0.02135	0.01381	−1.55
Market price probs. × *SAD*							0.00722	0.00418	1.73
Log-likelihood	−130066.63			−130062.20			−130052.30		
Pseudo-R^2	0.1627			0.1628			0.1629		

Notes:
* Indicates significance at the 5% level.
** Indicates significance at the 1% level.
Tests for multicollinearity among the explanatory variables reveal variance inflation factors (VIF) and tolerances well within acceptable limits (max values of 1.31 and 0.76, respectively; see Blaikie, 2003.

mood misattribution bias that is induced by environmental conditions.

As a consequence, we examined the degree of improvement in forecasting accuracy that can be achieved by correcting for the mood misattribution bias. We achieved this by using the three models presented in Table 4 (i.e., estimated using the training races) for forecasting winning probabilities for the races during the holdout period (01 Jan, 2014–31 Dec, 2016). These probabilities were used as inputs for developing a 0.5 Kelly betting strategy (with an initial wealth of $1000) without reinvestment of the

Table 5
Rates of return achieved using a 0.5 Kelly betting strategy using winning probabilities forecast for the holdout races (1 Jan, 2014 –31 Dec, 2016) by the benchmark CL, preference CL and EC CL models, with coefficients estimated using the training data (1 Jan, 2002 –31 Dec, 2013).

Odds probabilities:	No. bets	No. races with profit	Amt. bet ($)	Profit ($)	Rate of ret. without reinvestment %
Benchmark CL					
>50%	449	256	8156.74	−174.46	−2.14%
≤50% and >20%	606	214	5675.17	−775.54	−13.67%
≤20%	38	6	170.12	−36.49	−21.45%
Total	**1093**	**476**	**14 002.04**	**−986.49**	**−7.05%**
Preference CL					
>50%	449	260	9474.32	−88.33	−0.93%
≤50% and >20%	761	266	8103.76	259.43	3.20%
≤20%	61	5	326.99	−102.46	−31.33%
Total	**1271**	**531**	**17 905.07**	**68.64**	**0.38%**
EC CL					
>50%	451	258	11 566.12	110.54	0.96%
≤50% and >20%	841	295	10 928.79	551.31	5.04%
≤20%	115	11	781.09	−326.08	−41.75%
Total	**1404**	**564**	**23 276.00**	**335.77**	**1.44%**

winnings (as was outlined in Section 4.5). The results of applying this strategy, using probabilities derived from the 'benchmark CL', 'preference CL' and 'EC CL' models, are presented in Table 5.

The overall rate of return (ROR) values for the 'benchmark CL', 'preference CL' and 'EC CL' models are −7.05%, 0.38% and 1.44% respectively. Clearly, the ROR of the 'preference CL' model is substantially larger than that obtained for the 'benchmark CL' (a difference in ROR of 7.43%). In addition, the difference in ROR between the 'EC CL' and 'preference CL' is 1.06%, representing a proportional improvement of 279%.

When comparing the RORs from the strategies displayed in Table 5 for bets in different odds ranges, it can be observed that the 'EC CL' has the best economic performance for bets at odds probabilities of greater than 20%. For example, the 'EC CL' is the only strategy that achieves a positive ROR at odds probabilities greater than 50% (i.e., ROR of 0.96%, compared to RORs of −2.14% and −0.93% for the 'benchmark CL' and 'preference CL', respectively), as well as achieving the highest ROR for odds probabilities between 20% and 50% (i.e., ROR of 5.04% compared to RORs of −13.67% and 3.20% for the 'benchmark CL' and 'preference CL', respectively). Conversely, the 'EC CL' has the poorest economic performance for odds probabilities of less than or equal to 20% (i.e., ROR of −41.75%, compared to RORs of −21.45% and −31.33% for the 'benchmark CL' and 'preference CL', respectively). However, the small numbers of bets related to some of these categories prevent us from drawing any firm conclusions in relation to these different odds probability ranges.

Next, a bootstrap procedure was used to determine whether the differences in the returns between the three strategies ('benchmark CL', the 'preference variable CL' and the 'EC CL') were significant. This was achieved by drawing random samples of races from the holdout period, with replacement, where each sample consists of the same number of observations as in the holdout period. This procedure was repeated 1000 times. Then, we determined the return from a 50% Kelly betting strategy for each of the resulting 1000 samples based on winning probabilities forecast by the 'benchmark CL', the 'preference variable CL' and the 'EC CL'. The resulting distributions of returns were used to test whether the differences in the returns that could be achieved by each of these three strategies were statistically significant. In fact, t-tests showed that the differences in returns obtained from winning probability forecasts based on the 'benchmark CL' and 'preference CL' models and between the 'preference CL' and 'EC CL' models were significantly different at the 1% level ($t\,(1000) = 35.59$ and 3.45, respectively).

These results indicate that the model that accounts for the effect of EC on market prices ('EC CL') provides additional information over and above that incorporated in those models that simply include market prices ('benchmark CL') and the influence of EC on horse and jockey performances ('preference CL') (i.e., the returns that can be achieved using winning probability forecasts derived from the 'EC CL' model are significantly greater than those that are possible using either of the other models). These results support H2 and confirm the significance of the influence of EC on the forecast accuracy of probabilities derived from market prices.

We went on to investigate the influences of EC on the forecast accuracies of market prices. In particular, we examined which EC had the greatest impacts on the forecasting accuracies of odds-implied probabilities. We achieved this by examining the ROR from a betting strategy which employs probabilities that are corrected for the influence of particular individual EC conditions on the accuracy of the odds probabilities (i.e., using 'EC CL'), and comparing the returns achieved for this strategy under different EC. If, as was proposed in H1, good (bad) mood leads to worse (better) calibrated forecasts, then we expect larger (smaller) inaccuracies in the odds-implied probabilities (i.e., the betting strategy using the corrected odds-implied probabilities should achieve higher (lower) ROR) under EC associated with a more positive (negative) mood. In conducting this analysis, we categorised our EC variables into those associated with positive and negative mood on the basis of the literature referred to in footnote 2. The results of this analysis are displayed in Table 6.

The results demonstrate that substantially greater RORs are achieved under EC that have been shown to induce more positive moods. Interestingly, positive RORs were achieved under all EC associated with positive mood and

Table 6
Rates of return achieved for the 0.5 Kelly betting strategy based on estimates from the EC CL model for different EC.

	Environmental condition	Amt. bet ($)	Profit ($)	Rate of ret. without reinvestment %
Conditions associated with good mood	*No full moon*	20 119.15	397.54	1.98%
	No geomagnetic storms	9792.47	645.01	6.59%
	Not SAD months	16 060.27	1041.80	6.49%
	No rain	21 765.38	617.50	2.84%
	Clear skies (cloud cover $\leq$ 3)	8034.20	237.39	2.95%
	Positive deseasonalized atmospheric pressure	10 422.00	481.41	4.62%
	Low wind conditions ($\leq$ 5 km/h)	2412.81	487.54	20.21%
	Good air quality (air quality index $\leq$ 3)	20 956.58	434.32	2.07%
	Positive deseasonalized temperature	12 092.96	361.40	2.99%
	Lower humidity ($\leq$ 60%)	7556.08	375.53	4.97%
Conditions associated with bad mood	*Full moon*	3156.85	−61.76	−1.96%
	Geomagnetic storms	13 483.53	−309.23	−2.29%
	SAD months	7215.72	−706.02	−9.78%
	Raining days	1510.62	−281.72	−1865%
	Cloudy days (cloud cover $\geq$ 7)	10 533.41	−517.15	−4.91%
	Negative deseasonalized atmospheric pressure	12 853.99	−145.63	−1.13%
	Windy (>5 km/h)	20 863.18	−151.76	−0.73%
	Poor air quality (air quality index $\geq$4)	2319.41	−98.54	−4.25%
	Negative deseasonalized temperature	11 183.03	−25.62	−0.23%
	Higher humidity (>60%)	15 719.92	−39.75	−0.25%

negative RORs were achieved under all EC associated with negative mood. Thus, these results provide further evidence of the influence of EC on prediction calibration. In particular, they suggest that the EC associated with good mood have a negative influence on the forecasting accuracy of market prices, providing further evidence supporting H1.

6. Conclusion

Our principal aim was to identify the extent to which the forecast probabilities derived from market prices in prediction markets are affected by environmental conditions. The results offer an interesting insight into the degree to which EC may affect individuals' abilities to process information concerning future states of the world.

The results suggest that decision-makers in the prediction markets studied are skilful at making probabilistic forecasts of event outcomes. However, under certain EC, the market prices deviate from rational asset pricing, thus leading to less accurate probability forecasts.

To the best of our knowledge, this is the first paper to investigate the influence of EC-induced mood on prices in a prediction market, where the EC themselves are also important factors when estimating the fundamental performances of the contracts traded. For instance, the nature of the prediction market studied means that participants are 'nudged' to consider EC, as they influence the fundamental performances of horses and jockeys. Studies in psychology have demonstrated that individuals' decisions are less likely to be influenced by the effects of EC when they are nudged about the EC (Schwarz & Clore, 1983). However, despite the fact that one might expect bettors in horserace betting markets to be alert to the effects of EC, we observe a significant influence of EC on market prices. This suggests that the effects of EC on market prices are likely to be even more significant in other prediction markets in which participants do not feel the need to consider EC actively.

The only logical way in which the EC examined could influence prices in these markets is through their effects on the performances of horses and jockeys. However, we controlled for the influences of EC on the performances of both horses and jockeys. As a consequence, our results strongly suggest that a mood misattribution bias has a negative influence on the accuracy of forecasts derived from final market prices.

We have demonstrated a systematic link between the current EC and the forecast accuracy of market prices, as measured by the Brier score. Specifically, we have showed that the *warmer, sunnier and drier weather* and *deteriorating weather* components and *SAD* all had statistically significant influences on the Brier score. This is a surprising result, especially considering that the model estimated included eight control variables that accounted for factors that previous research has shown to influence the forecast accuracy, and that the model controlled for the influence of the EC on the performances of horses and jockeys. These results provide a clear indication of the presence of an EC-induced mood misattribution bias, and suggest the possibility of improving the probability estimates derived from market prices by correcting for this bias.

We observed that although the winning probability forecasts derived from market prices were highly predictive, they do not account for the influences of EC on the performances of horses and jockeys fully. In addition, even after controlling for such effects, misattribution bias still led to sub-optimal probability estimates. We demonstrated that it is possible to improve the forecast accuracy of market price probabilities significantly by correcting for the misattribution bias detected, and that this could lead to substantial improvements in the rate of return derived from a betting strategy based on the adjusted winning probability forecasts.

There is no appealing explanation from traditional EMH as to why the EC studied here should have any effect on market prices, other than through their influence on the performances of horses and jockeys. However, our results

are consistent with the psychological literature, which suggests that EC influence mood and judgements about the future, thereby affecting the quality of the forecasts about future states of the world.

Our study suggests that the participants' judgements are influenced by EC even in prediction markets that consist of skilled and experienced participants, leading to less accurate forecasts. We show that significantly better forecasts can be achieved by correcting for this phenomenon, and that these improvements have substantial economic value.

In summary, we arrive at the clear conclusion that when the purpose of a prediction market is to derive accurate probabilistic estimates from final contract prices, the forecast accuracy can be improved substantially by understanding and correcting for situations in which prediction markets systematically underperform. Thus, the implications of this research are far-reaching.

The effective use of prediction markets has helped many organisations to predict uncertain and complex outcomes, such as the results of political events, the demand for products, production costs across business units, and the likelihood of success of design innovations (Healy, Linardi, Lowery, & Ledyard, 2010; Soukhoroukova et al., 2012). Our results suggest that the underlying value of prediction markets to organisations can be improved greatly by identifying and correcting for conditions under which individuals systematically make sub-optimal estimations about future states of the world.

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